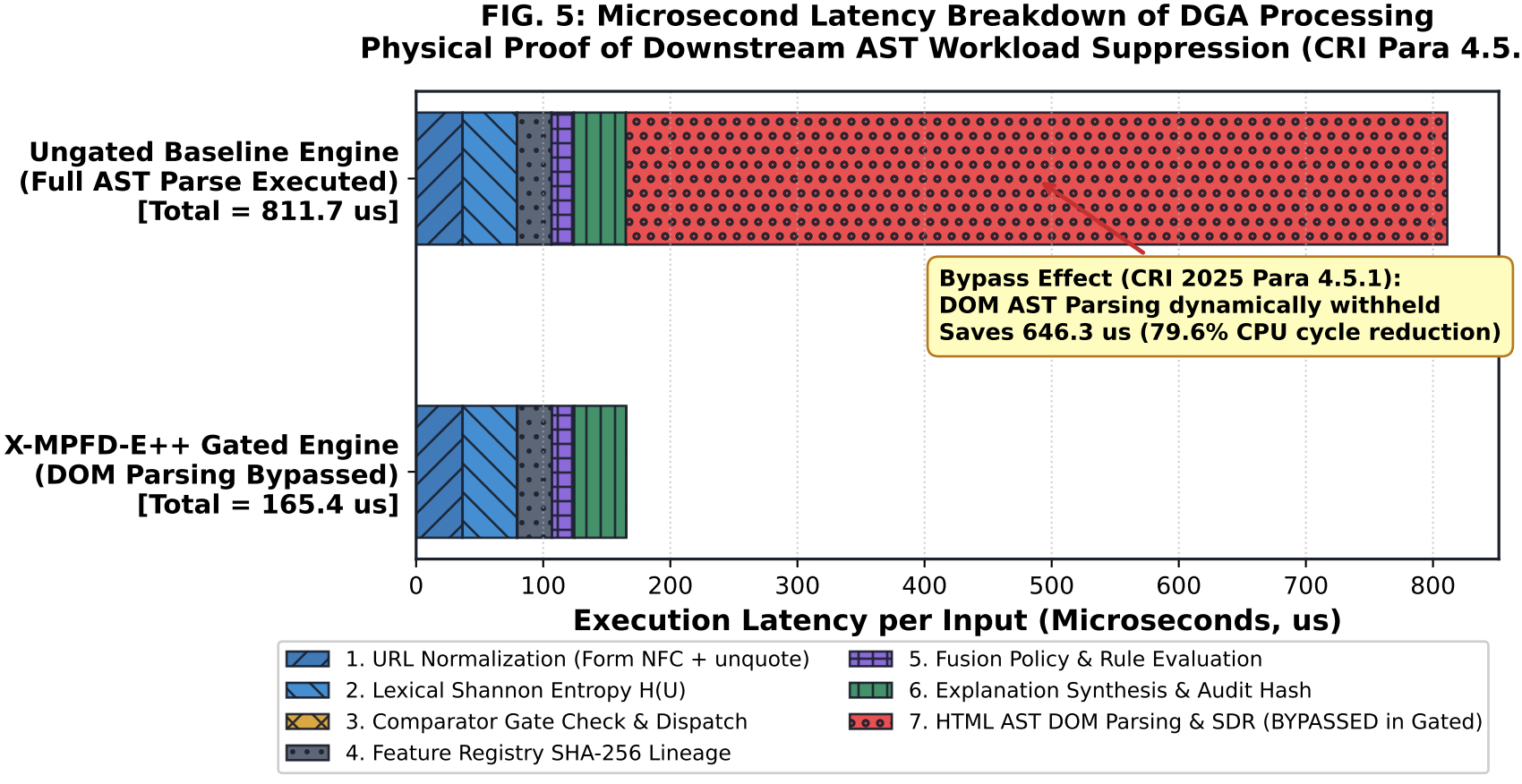

FIG. 5: Microsecond Latency Breakdown of DGA Processing
Physical Proof of Downstream AST Workload Suppression (CRI Para 4.5.
Ungated Baseline Engine
(Full AST Parse Executed)
[Total = 811.7 us]
X-MPFD-E++ Gated Engine
(DOM Parsing Bypassed)
[Total = 165.4 us]
Bypass Effect (CRI 2025 Para 4.5.1):
DOM AST Parsing dynamically withheld
Saves 646.3 us (79.6% CPU cycle reduction)
0
100
200
300
400
500
600
700
800
Execution Latency per Input (Microseconds, us)
1. URL Normalization (Form NFC + unquote)
2. Lexical Shannon Entropy H(U)
3. Comparator Gate Check & Dispatch
4. Feature Registry SHA-256 Lineage
5. Fusion Policy & Rule Evaluation
6. Explanation Synthesis & Audit Hash
7. HTML AST DOM Parsing & SDR (BYPASSED in Gated)